

GD HW2

1) sequence $\{x_i\}$ is p -periodic if $\exists N, p \in \mathbb{N}$ such that

$$x_{N+j} = x_{N+j+p} \quad \forall j \in \mathbb{N}$$

ordinary GD? update $x_{k+1} = x_k - \gamma \nabla f(x_k)$

strictly convex $f(x) = x^2$

parameters $x_1 = 1, \gamma = 1$

sequence $x_1 = 1$

$$x_2 = 1 - 2 = -1$$

$$x_3 = -1 + 2 = 1$$

... is non-constant, 2-periodic

polyak GD? update $x_{k+1} = x_k - \gamma \nabla f(x_k) + \mu(x_k - x_{k-1})$

$$\text{(notes)} f'(x) = \begin{cases} 25x, & x < 1 \\ x+24, & 1 \leq x \leq 2 \\ 25x-24, & x > 2 \end{cases}$$

f' continuous and increasing $\rightarrow f$ strictly convex

parameters $\gamma = \frac{1}{9}, \mu = \frac{4}{9}$

initialization + sequence

$$x_1 = 0.65 = w_1$$

$$x_2 = -1.8 = w_2$$

$$x_3 = 2.12 = w_3$$

$$x_4 = w_1$$

is 3-periodic

Nesterov GD? update $x_{k+1} = x_k + \frac{\mu(x_k - x_{k-1})}{1 - \gamma\mu} - \gamma f'(x_k + \frac{\mu(x_k - x_{k-1})}{1 - \gamma\mu})$

$$\text{construct piecewise } f'(x) = \begin{cases} x-1, & x \leq 0 \\ -1 + \frac{1}{3}x, & 0 < x \leq \frac{3}{2} \\ -\frac{1}{2} + 3(x-5), & x > \frac{3}{2} \end{cases}$$

also f strictly convex

parameters $\gamma = 1, \mu = \frac{1}{2}$

initialization + cycle $x_0 = 0$

$$x_1 = 0$$

$$x_2 = 1$$

$$x_3 = 2$$

$$x_4 = 0$$

$$x_5 = 1$$

... also 3-periodic

$$2) f(x, y, z) = x^2 + 2y^2 - 2yz + 4z^2 + 3x - 4y + 5z$$

theorem 5.2: optimal Polyak GD parameters for quadratic function

$$\gamma = \frac{4}{(\sqrt{\alpha} + \sqrt{\beta})^2} \quad \sqrt{\mu} = \frac{\sqrt{\beta} - \sqrt{\alpha}}{\sqrt{\beta} + \sqrt{\alpha}} *$$

$$\text{Hessian } \nabla^2 f = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & -2 \\ 0 & -2 & 8 \end{bmatrix} \rightarrow \lambda_{2,3} = \frac{a+d}{2} \pm \sqrt{\left(\frac{a-d}{2}\right)^2 + b^2}$$

$\downarrow$
 $\lambda_1 = 2$

$$\alpha = \lambda_{\min} = 2$$

$$\beta = \lambda_{\max} = 6 + 2\sqrt{2}$$

plug in *

$$\gamma = 0.208$$
$$\mu = \underline{\underline{0.126}}$$

which also gives us some convergence guarantee